

Estimates of the economic return to schooling for 28 countries[☆]

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Received 20 July 1999; received in revised form 4 July 2001; accepted 30 August 2001

Abstract

The economic returns to schooling are estimated using comparable microdata in 28 countries, worldwide. Considerable variation in rates of return is found across countries. There is no evidence for a worldwide rising rate of return to education from 1985 through 1995. Indeed, the worldwide rate of return declines slightly over this period. In general, instrumental-variable estimates (using spouse's and parents' schooling as determinants of schooling) are over 20% higher than ordinary-least-squares estimates. © 2002 Elsevier Science B.V. All rights reserved.

JEL classification: Returns to education; Instrumental variables

1. Introduction

This paper estimates the rates of return to education in 28 countries using comparable microdata from 1985 through 1995. Our aim is to investigate how widespread are two important features of the recent, largely UK and US, literature: the rising rate of return to education, and the finding that least-squares estimates are biased downwards rather than, as had commonly been thought more plausible, upwards.

[☆] We are grateful to the ESRC Data Archive at the University of Essex for supplying the data used in this analysis, to the Department for Education and Skills for funding the research and to an anonymous referee. The data used in the analysis and further results can be obtained from Professor Ian Walker.

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This paper complements Psacharopoulos's (1994) summary of the evidence on rates of return to schooling across countries. Our study, however, has the added benefit of strictly comparable data. The paper also complements the recent research surveyed by Card (1999) that is concerned with the sensitivity of instrumental-variable estimates to the choice of instruments since here we are able to use consistent instruments across countries.

We begin by presenting conventional estimates of Mincerian human capital wage functions: that is, OLS estimates of rates of return to schooling.¹ We find a great deal of heterogeneity in the rate-of-return estimates across countries. The estimates range from 1.9% for women in the Netherlands, to 19.2% for women in the Philippines. However, we find it difficult to explain much of this cross-country variation. There is tenuous evidence that the rate of return declines with average educational attainment (i.e., diminishing returns to schooling), per capita income, and, surprisingly, relative spending on education. There is somewhat stronger evidence that the estimated rate of return is higher when wages are measured before taxes rather than after taxes (thus suggesting that, in general, the structures of labour-income taxes are progressive).

We have multiple cross-sections of data for most of the 28 countries, which allows us to investigate how the returns vary over time. Contrary to evidence from data from the US in the 1980s, there is essentially no evidence of a rising rate of return. In most countries there is no significant trend in the rate of return to education, and, overall, there is evidence of a slight decline in the worldwide rate of return over the 1985–1995 period, particularly for women.

Finally, we address the problem of the potential endogeneity of schooling by presenting instrumental-variable estimates of the rate of return. Consistent with most previous evidence, the IV estimates are substantially greater than the OLS estimates; that is, OLS estimates appear to be biased downward significantly. We have data on spouse's education in 10 of the 28 countries. Thus, for couples we can exploit the very strong correlation between spouses' education levels and the lack of correlation between the wage of one spouse and the education of the other. Using spouse's education to instrument for observed schooling yields estimates of the rate of return that, on average, are a little over 20% higher than the corresponding OLS estimates for couples. We also have data on father's education in nine countries, and on mother's education in eight countries. While the use of education levels of the father and the mother to instrument for schooling is questionable on the grounds that it seems likely that they affect wages conditional on education as well as education, we find that this yields the same conclusion: OLS estimates appear to be biased downward by at least 20%.

2. Data

We use International Social Survey Programme data, 1985–1995. These data were collected in each of a large number of countries using a common questionnaire. While the focus

¹ We also used maximum-likelihood estimates for the cases when wages are only observed to fall within particular intervals, but little difference is found between the OLS and the ML interval estimates. These are available on request.

Table 1
Summary statistics

Country	No. of years	Males			Females		
		Sample size	Mean schooling	Standard deviation	Sample size	Mean schooling	Standard deviation
USA	11	3126	13.64	2.84	3304	13.61	2.47
Great Britain	11	2640	11.47	1.49	2583	11.46	1.46
West Germany	9	3130	10.60	3.18	2047	10.48	2.65
Russia	5	2392	13.20	3.34	2671	13.06	3.12
Norway	7	2568	12.63	2.97	2332	12.46	3.00
Australia	6	2865	11.74	2.76	1964	11.69	2.69
Netherlands	7	2179	13.40	3.92	1320	13.58	3.55
Austria	8	1645	11.09	2.66	1226	10.79	2.42
Poland	5	1414	11.08	2.65	1379	11.85	2.68
East Germany	5	1158	10.89	2.93	1096	10.65	2.52
New Zealand	5	1033	12.79	3.20	1122	12.78	2.99
Italy	6	1245	11.97	3.98	790	12.24	3.87
Ireland	6	1085	12.18	3.12	859	12.76	2.71
Japan	3	772	13.10	2.55	618	12.22	1.89
Hungary	3	650	11.55	2.85	661	11.56	2.89
N. Ireland	5	659	11.70	1.50	631	11.78	1.39
Sweden	2	558	12.07	3.32	641	12.08	3.13
Slovenia	3	573	11.13	2.76	578	11.42	2.72
Israel	2	469	12.97	3.16	559	13.46	2.72
Czech Rep.	2	433	13.24	2.81	360	13.19	2.68
Bulgaria	2	374	11.55	3.09	375	12.14	2.91
Slovak Rep.	1	355	12.63	2.68	298	12.78	2.43
Canada	1	270	15.69	3.45	318	15.85	3.40
Czechoslovakia	1	301	12.85	2.60	272	12.57	3.17
Spain	2	277	10.98	4.74	141	12.38	4.72
Switzerland	1	285	11.16	3.79	126	10.66	3.15
Latvia	1	141	12.44	2.84	190	13.45	2.82
Philippines	1	197	9.80	4.03	86	11.66	4.03
Pooled	11	32,794	12.13	3.19	28,547	12.25	2.94

of the questions varied from year to year, there is a common core of questions that includes individual earnings, education, marital status and so forth that is available every year. Our sample is of employed individuals aged 21–59 in the year of interview. National cross-sectional surveys are pooled across (up to) 11 years from 1985 to 1995. Individuals self-employed, in school, or retired are not included. Table 1 presents some summary statistics.

3. OLS results

The dependent variable is the logarithm of hourly earnings (y_i), computed as weekly earnings divided by the number of hours usually worked per week. In some of the countries in some years, however, the weekly earnings variable is only observed to fall

within particular intervals on a continuous scale. In these cases we use interval midpoints for weekly earnings.²

The estimation does not correct for selectivity into employment. Unfortunately, there is no suitable variable that is correlated with participation that could be justifiably excluded from the wage equation.³

Initial estimates for males and females in each the country are obtained using the conventional Mincer (1974) model of earnings (the human capital earnings function), which has log wage rates determined by years of schooling, age or experience and other explanatory variables:

$$y_i = \mathbf{X}_i' \alpha + \beta S_i + u_i, \quad (1)$$

where y_i is the log of hourly wages, S_i is years of schooling and $\mathbf{X}_i$ is a vector of observed exogenous explanatory variables including, where appropriate, country and year fixed effects. β is interpreted as the rate of return to schooling; namely, the percentage change in wages due to an additional year of schooling.

OLS estimates of β for males and females in the 28 countries being studied are presented in Table 2.⁴ Pooling the samples suggests a worldwide OLS estimate of the rate of return to schooling in the order of 4.8% for men, and 5.7% for women. Both the magnitude and the gender differential are consistent with the previous literature. Other international estimates can be found in Psacharopoulos (1994).

4. Cross-country heterogeneity

Bearing in mind that the estimates are generated using uniform procedures and broadly comparable data across countries, the cross-country variation in the estimated rates of return is quite striking. The highest estimate (19.2% for females in the Philippines) is 10 times higher than the lowest estimate (1.9% for females in the Netherlands). Furthermore, the rates of return are typically estimated with considerable precision. But what may be the most puzzling aspect of the cross-country heterogeneity is the lack of obvious explanations for it. Few general patterns are apparent in the cross-country variation. It appears that the returns are generally higher outside Continental Europe, but this seems to be the only clear pattern.

There is little apparent correlation between the rate of return and per capita income. This is revealed in Fig. 1, which plots the estimated return in each country (for men and women separately) against its per capita income.⁵ The highest returns are found in

² We extrapolate values for the top open-ended group. Our estimates, however, are not sensitive to this choice.

³ Recent work, for example by Dearden (1998), shows that failing to adjust for selection has little or no impact on estimates of schooling returns.

⁴ In many of our countries the earnings data are reported in intervals. ML estimates using Stewart's (1983) interval-regression technique are available on request. The grouped nature of the dependent variable produces only minor differences in the estimates. As little is gained using maximum likelihood we simply use the midpoint of the data.

⁵ To be precise, per capita income is average real GNP per capita, PPP, over the 10-year period 1985–1994 from the World Bank's International Comparison Programme database.

Table 2
OLS estimates

Country	Males		Females	
USA	0.074	<i>0.004</i>	0.096	<i>0.005</i>
Great Britain	0.127	<i>0.006</i>	0.130	<i>0.006</i>
West Germany	0.036	<i>0.002</i>	0.043	<i>0.004</i>
Russia	0.044	<i>0.004</i>	0.053	<i>0.004</i>
Norway	0.023	<i>0.002</i>	0.025	<i>0.003</i>
Australia	0.051	<i>0.004</i>	0.052	<i>0.006</i>
Netherlands	0.031	<i>0.002</i>	0.019	<i>0.004</i>
Austria	0.038	<i>0.004</i>	0.064	<i>0.006</i>
Poland	0.073	<i>0.005</i>	0.100	<i>0.005</i>
East Germany	0.026	<i>0.003</i>	0.045	<i>0.004</i>
New Zealand	0.033	<i>0.004</i>	0.029	<i>0.005</i>
Italy	0.037	<i>0.003</i>	0.053	<i>0.005</i>
Ireland	0.085	<i>0.006</i>	0.090	<i>0.008</i>
Japan	0.075	<i>0.007</i>	0.094	<i>0.014</i>
Hungary	0.075	<i>0.007</i>	0.077	<i>0.006</i>
N. Ireland	0.174	<i>0.011</i>	0.146	<i>0.011</i>
Sweden	0.024	<i>0.004</i>	0.033	<i>0.005</i>
Slovenia	0.080	<i>0.007</i>	0.101	<i>0.007</i>
Israel	0.053	<i>0.007</i>	0.061	<i>0.008</i>
Czech Rep.	0.035	<i>0.007</i>	0.043	<i>0.007</i>
Bulgaria	0.040	<i>0.009</i>	0.057	<i>0.010</i>
Slovak Rep.	0.052	<i>0.012</i>	0.064	<i>0.009</i>
Canada	0.038	<i>0.008</i>	0.045	<i>0.008</i>
Czechoslovakia	0.031	<i>0.010</i>	0.036	<i>0.007</i>
Spain	0.046	<i>0.005</i>	0.038	<i>0.010</i>
Switzerland	0.045	<i>0.007</i>	0.048	<i>0.012</i>
Latvia	0.067	<i>0.020</i>	0.078	<i>0.014</i>
Philippines	0.113	<i>0.015</i>	0.192	<i>0.030</i>
Pooled	0.048	<i>0.001</i>	0.057	<i>0.001</i>

Robust standard errors are in italics. The estimating equations include year dummies, union status, marital status, age and age squared and, in the case of the aggregate equation, country-year dummies.

countries with incomes that are relatively high (USA and Japan) and relatively low (Philippines), as well as in-between (Northern Ireland, GB, Slovenia and Poland). Also, as shown in Fig. 2, there is little apparent pattern between the rate of return and average educational attainment in our samples. Similarly, as illustrated in Fig. 3, there is little relationship apparent between the return and the percentage of GNP spent on education.⁶

There are, however, two differences in the data across countries that appear to be important in explaining some of the differences in returns to schooling. Measured schooling is truncated between 10 and 14 years of schooling in Great Britain and Northern Ireland, and these countries have very high estimates of the rate of return. Given

⁶ To be more precise, educational spending relative to GNP is the UNESCO's average over the 10-year period 1985–1994.

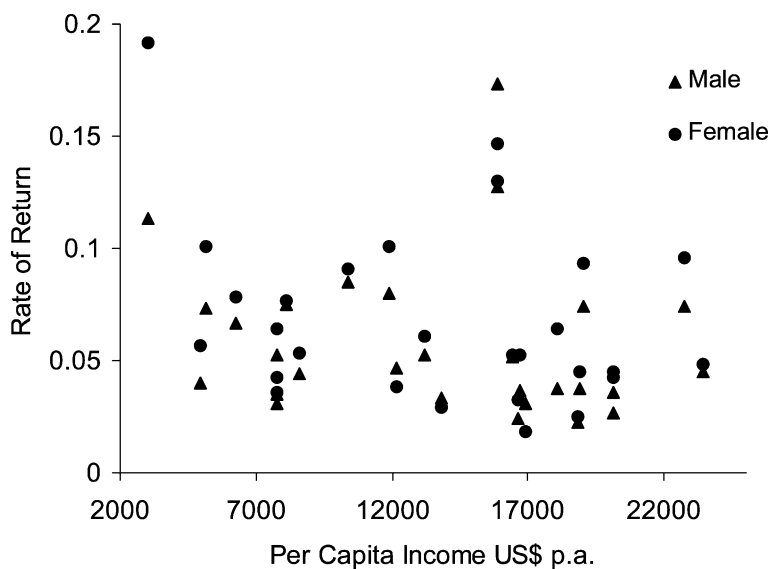


Fig. 1. Cross-country rates of return and per capita income.

that advanced degrees should lead to higher earnings, truncating measured education at 14 years probably upwardly biases these estimated rates of return. Also, earnings are measured before taxes in some of the countries (USA, GB, Norway, Australia, New

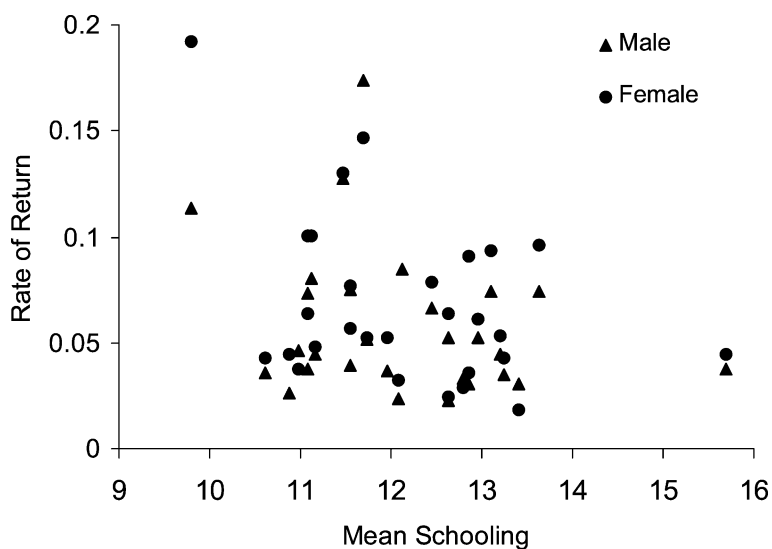


Fig. 2. Cross-country rates of return and schooling.

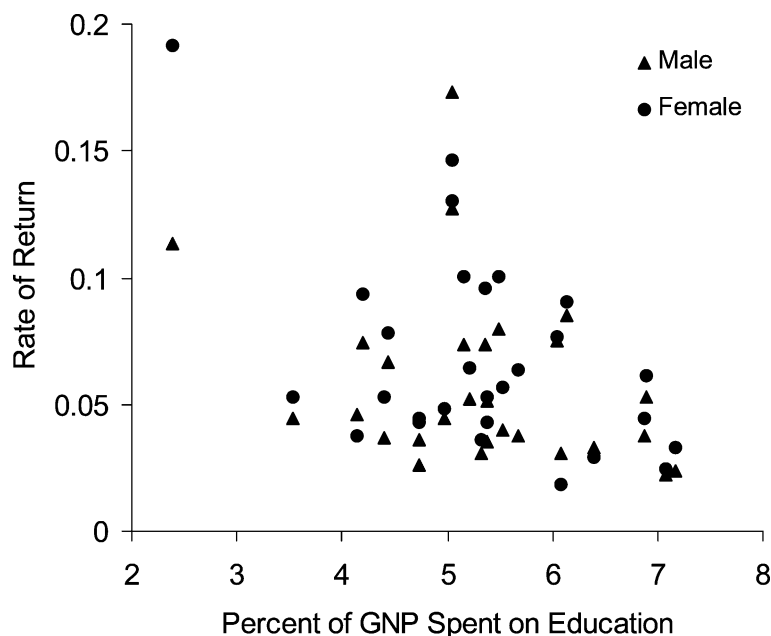


Fig. 3. Cross-country rates of return and education spending.

Zealand, Japan, Northern Ireland and Sweden) and after taxes in some of the countries (East and West Germany, Netherlands, Austria, Poland, Italy, Hungary, Slovenia, Czechoslovakia and Czech Republic, Switzerland and Latvia). There are also several countries where the documentation is insufficient to determine if wages are measured before or after taxes (Russia, Israel, Bulgaria, Slovak Republic, Canada, Spain and Philippines), and Ireland measures earnings before taxes in three years, and after taxes in three years. If taxes on labour income are linear, then this distinction should not affect the estimated rate of return to schooling. If, however, the tax structures are progressive (and the incidence of taxes on labour income falls on workers), then the estimated rates of return should be lower when earnings are measured net of taxes. Indeed, the after-tax estimated rates of return are generally lower than the before-tax estimated returns. This is illustrated in Fig. 4.

We attempt to quantify these observations by estimating the correlation between the estimated rates of return and various factors that could reasonably create cross-country differences. These correlation coefficients are reported in Table 3. Surprisingly, the higher return for women is not statistically significant (although the difference is significant when the samples are pooled across countries). The negative correlation between the rates of return and per capita income is marginally significant. But the majority of this correlation comes from one outlier (Philippines). Similarly, there is a significant negative correlation between the return and the mean level of education, but its significance does not remain if the Philippine estimates are not included. There is also a significant negative correlation between the return and relative spending on

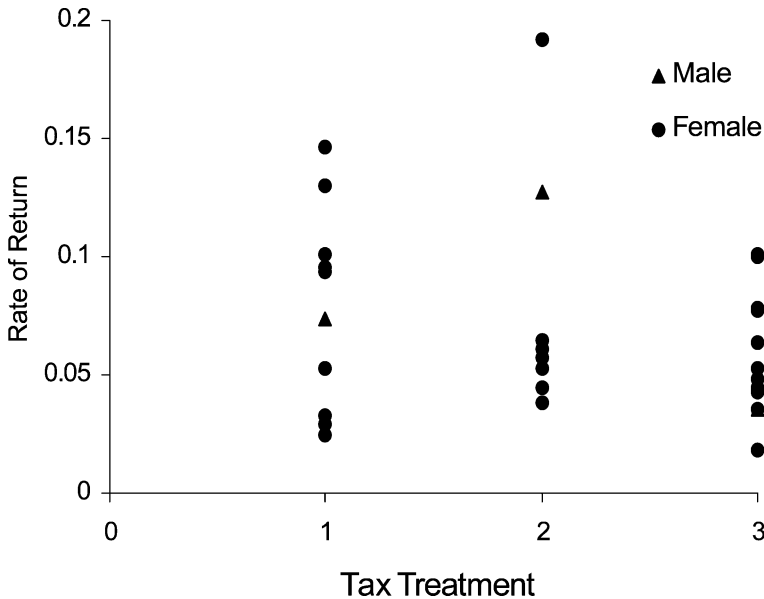


Fig. 4. Cross-country rates of return and tax status. 1 = before tax, 2 = unknown, 3 = after tax.

education, but about half of this correlation is removed if the Philippine estimates are not included. Estimates from before-tax wages are statistically higher than those from after-tax wages. Naturally, the dummy variable for the two countries with truncated values for schooling is very highly correlated with the estimated rate of return.⁷

5. The trend in the rate of return

Some recent studies using US data suggest that the rate of return to schooling is increasing over time, possibly due to increased returns to ability.⁸ By including a trend interaction with schooling we estimate the how the return to schooling has grown over time, on average, over the years in the data. Table 4 shows these results for the countries with a minimum of five years of data. In most countries there are no significant

⁷ Following the recent tests for publication bias in the literature by Ashenfelter et al. (1999), we also estimated the correlation between the estimated rates of return and their standard errors. Surprisingly, there is a positive correlation with the standard error in our results (although this correlation is only marginally significant if the Philippine estimates are excluded), while there is essentially no correlation between the coefficient estimate and the sample size (Fig. 5). In meta-analyses this has been interpreted as evidence of publication bias, but it is difficult to conceive of how this could occur here where we apply uniform methods across countries. Moreover, given the precision of the estimates in Table 2 (every *t* ratio exceeds three), the marginal incentive for data mining is essentially zero.

⁸ For example, Blackburn and Neumark (1993), Murnane et al. (1995) and Cawley et al. (1995).

Table 3
Cross-country variation in rates of return

	Correlation coefficient	<i>P</i> -value
Female dummy	0.138	0.310
Per capita income	−0.212	0.085
Mean schooling	−0.317	0.017
School spending/GNP	−0.416	0.001
Dummy for before tax wages	0.237	0.074
Dummy for after tax wages	−0.222	0.093
Truncation dummy	0.611	0.000
Standard error	0.485	0.002
Sample size	−0.050	0.716

increases (or indeed decreases) in the returns to schooling, including the USA. When we include a quadratic trend, however, the US return for males rises initially before falling in the later years. Moreover, both the linear and quadratic trend coefficients are marginally significant for US men (although not for US women). Thus, the evidence is consistent with the previous US evidence, but it appears that the trend was reversed in first half of the 1990s.

In general, though, for males there is little evidence that the return to education is either increasing or decreasing appreciably worldwide. There are equal numbers of negative and positive trend coefficients for men, and most are not significantly different from zero.

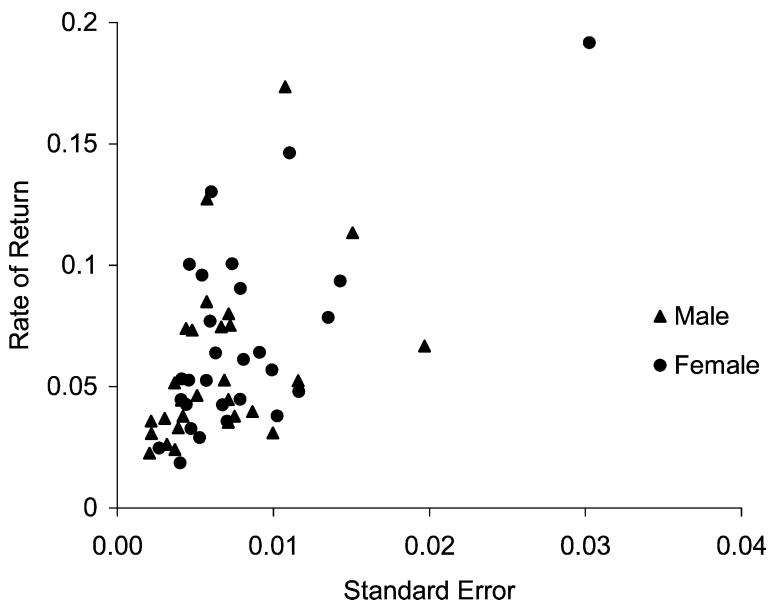


Fig. 5. Cross-country OLS rates of return and their standard errors.

Table 4
OLS estimates of the trend in the rate of return

Country	Males				Females			
	Initial rate of return		Trend		Initial Rate of Return		Trend	
USA	0.0742	<i>0.0078</i>	– 0.0001	<i>0.0014</i>	0.0963	<i>0.0110</i>	– 0.0001	<i>0.0018</i>
Great Britain	0.1158	<i>0.0099</i>	0.0026	<i>0.0019</i>	0.1501	<i>0.0101</i>	– 0.0042	<i>0.0018</i>
West Germany	0.0455	<i>0.0042</i>	– 0.0020	<i>0.0007</i>	0.0546	<i>0.0089</i>	– 0.0023	<i>0.0014</i>
Russia	0.0260	<i>0.0049</i>	0.0137	<i>0.0032</i>	0.0374	<i>0.0045</i>	0.0118	<i>0.0030</i>
Norway	0.0388	<i>0.0047</i>	– 0.0048	<i>0.0012</i>	0.0344	<i>0.0057</i>	– 0.0028	<i>0.0015</i>
Australia	0.0612	<i>0.0067</i>	– 0.0022	<i>0.0012</i>	0.0863	<i>0.0115</i>	– 0.0069	<i>0.0018</i>
Netherlands	0.0279	<i>0.0033</i>	0.0008	<i>0.0008</i>	0.0475	<i>0.0070</i>	– 0.0072	<i>0.0015</i>
Austria	0.0397	<i>0.0089</i>	– 0.0003	<i>0.0013</i>	0.0660	<i>0.0127</i>	– 0.0004	<i>0.0017</i>
Poland	0.0660	<i>0.0080</i>	0.0035	<i>0.0034</i>	0.0983	<i>0.0085</i>	0.0010	<i>0.0031</i>
East Germany	0.0199	<i>0.0044</i>	0.0037	<i>0.0024</i>	0.0391	<i>0.0052</i>	0.0032	<i>0.0025</i>
New Zealand	0.0403	<i>0.0066</i>	– 0.0042	<i>0.0031</i>	0.0220	<i>0.0100</i>	0.0041	<i>0.0043</i>
Italy	0.0350	<i>0.0089</i>	0.0004	<i>0.0017</i>	0.0737	<i>0.0115</i>	– 0.0046	<i>0.0023</i>
Ireland	0.0734	<i>0.0099</i>	0.0035	<i>0.0021</i>	0.0899	<i>0.0145</i>	0.0002	<i>0.0031</i>
N. Ireland	0.1960	<i>0.0171</i>	– 0.0091	<i>0.0056</i>	0.1417	<i>0.0162</i>	0.0020	<i>0.0060</i>
Pooled	0.0532	<i>0.0023</i>	– 0.0008	<i>0.0003</i>	0.0766	<i>0.0033</i>	– 0.0028	<i>0.0005</i>

Robust standard errors are in italics. The estimating equations include year dummies, union status, marital status, age and age squared and, in the case of the aggregate equation, country-year dummies.

When the samples are pooled, however, the overall trend in the return to schooling is slightly downward for men over the 1985–1995 period.⁹ For women, there is slightly stronger evidence of a declining rate of return. But even for women, the evidence is far from uniform across countries. Most trend coefficients for females are insignificant, and there are almost equal numbers of positive and negative coefficients.

6. Instrumental-variables results

It has become well known that the OLS estimate of the rate of return to education is unbiased only if measured schooling is exogenous. Endogeneity arising from measurement error in S is generally thought to bias the estimate of β towards zero, although this effect is believed to be small because the reliability of schooling data is typically quite high. Secondly, endogeneity can arise because of omitted ability. That is, the return coefficient β is biased (upwards) because chosen schooling levels are (positively) correlated with omitted ability, and ability is (positively) correlated with the wage rate. On the other hand, Card (1999) and others¹⁰ have argued that OLS estimates of β are biased downwards because individuals with high discount rates choose low levels of schooling, which have a higher marginal rate of return. Most of the recent studies reviewed

⁹ This is broadly consistent with previous evidence of a decreasing return in Europe; see for example, Goux and Maurin (1994) and Jarousse (1988).

¹⁰ Notably Lang (1993).

in Card (1999) suggest that OLS estimates of β are indeed biased downwards. Card reviewed only one non-US study, however. Thus it is of considerable interest to investigate the extent to which the conclusion is general.

To address endogeneity bias we need to instrument schooling by purging its correlation with unobservable influences on wages, using variables that are correlated with schooling but not with wage rates. Namely, the instrument needs to be orthogonal to the unobserved component of the wage equation. Such a joint model would be

$$y_i = \mathbf{X}_i' \alpha + \beta S_i + u_i \quad (2)$$

$$S_i = \mathbf{Z}_i' \delta + v_i$$

where $\mathbf{Z}_i$ is the vector of observed instrumental variables with the properties suggested above. Our data contains three potential instruments for S : spouse's education, father's education and mother's education.

It has been suggested in recent work (for example, Weiss, 1999) that marriage is subject to assortative mating. A common level of schooling is also more likely to lead to common experiences, and possibly common interests. Pencavel (1998) points out that, in US census data, husbands and wives have been becoming more similar in their schooling backgrounds. In 1990 there were 8.62 times as many couples with schooling differences of no

Table 5
IV Estimates using spouse's education to instrument for education

Country	Males		Females	
USA	0.084	<i>0.009</i>	0.116	<i>0.015</i>
	0.068	<i>0.005</i>	0.106	<i>0.009</i>
West Germany	0.042	<i>0.008</i>	0.069	<i>0.012</i>
	0.038	<i>0.004</i>	0.056	<i>0.009</i>
Australia	0.055	<i>0.011</i>	0.086	<i>0.021</i>
	0.055	<i>0.005</i>	0.060	<i>0.008</i>
Netherlands	0.048	<i>0.014</i>	0.053	<i>0.017</i>
	0.034	<i>0.006</i>	0.047	<i>0.011</i>
Poland	0.073	<i>0.009</i>	0.102	<i>0.014</i>
	0.071	<i>0.006</i>	0.106	<i>0.008</i>
East Germany	0.033	<i>0.010</i>	0.054	<i>0.019</i>
	0.029	<i>0.005</i>	0.039	<i>0.007</i>
Italy	0.075	<i>0.010</i>	0.113	<i>0.012</i>
	0.040	<i>0.004</i>	0.064	<i>0.006</i>
Ireland	0.088	<i>0.014</i>	0.132	<i>0.023</i>
	0.063	<i>0.009</i>	0.109	<i>0.013</i>
Hungary	0.081	<i>0.015</i>	0.103	<i>0.022</i>
	0.058	<i>0.011</i>	0.070	<i>0.011</i>
Czechoslovakia	0.043	<i>0.024</i>	0.046	<i>0.014</i>
	0.036	<i>0.011</i>	0.033	<i>0.008</i>
Weighted average	0.064	<i>0.011</i>	0.093	<i>0.017</i>
	0.053	<i>0.005</i>	0.076	<i>0.008</i>

Robust standard errors are in italics. The estimating equations include year dummies, union status, marital status, age and age squared. For comparison, the corresponding OLS estimates are shown in the smaller font underneath.

Table 6
IV Estimates using father's education to instrument for education

Country	Males		Females	
USA	0.106	<i>0.016</i>	0.136	<i>0.021</i>
	0.076	<i>0.006</i>	0.096	<i>0.007</i>
West Germany	0.042	<i>0.009</i>	0.056	<i>0.015</i>
	0.035	<i>0.003</i>	0.045	<i>0.007</i>
Australia	0.055	<i>0.011</i>	0.088	<i>0.033</i>
	0.048	<i>0.006</i>	0.035	<i>0.008</i>
Austria	0.053	<i>0.012</i>	0.078	<i>0.017</i>
	0.036	<i>0.006</i>	0.061	<i>0.009</i>
Poland	0.078	<i>0.013</i>	0.143	<i>0.018</i>
	0.072	<i>0.006</i>	0.099	<i>0.007</i>
East Germany	0.048	<i>0.019</i>	0.043	<i>0.028</i>
	0.028	<i>0.004</i>	0.033	<i>0.007</i>
Ireland	0.123	<i>0.025</i>	0.158	<i>0.043</i>
	0.078	<i>0.016</i>	0.129	<i>0.017</i>
Hungary	0.099	<i>0.027</i>	0.072	<i>0.023</i>
	0.057	<i>0.010</i>	0.073	<i>0.010</i>
Czechoslovakia	0.065	<i>0.024</i>	0.051	<i>0.014</i>
	0.033	<i>0.009</i>	0.036	<i>0.007</i>
Weighted average	0.072	<i>0.014</i>	0.103	<i>0.022</i>
	0.054	<i>0.006</i>	0.072	<i>0.008</i>

Robust standard errors are in italics. The estimating equations include year dummies, union status, marital status, age and age squared. For comparison, the corresponding OLS estimates are shown in the smaller font underneath.

more than one category than those with schooling differences of more than one category. We have data on spouse's education (as reported by the reference person) in 10 of the 28 countries.

We also have data on father's education (again, as reported by the reference person) in nine countries (eight of which are also in the group with data on spouse's education), and on mother's education in eight countries.

Table 5 presents IV estimates of β using spouse's education to instrument for S , together with the corresponding OLS estimates (i.e., for the subset of people recording spouse's education). Table 6 presents IV estimates using father's education as the instrument, along with the corresponding OLS estimates. IV estimates using mother's education as the instrument are given in Table 7.¹¹

The IV estimates shown in Tables 5–7 are consistent with the previous literature: namely, the OLS estimates of the rate of return to schooling are biased downward substantially. On average, the IV estimates using spouse's education are a little over 20% higher than the corresponding OLS estimates. Moreover, the IV estimates using either parent's education suggest an even larger difference (one third higher for men on average, and over 40% higher on average for women). In addition, because of the very strong

¹¹ IV estimates using both parents' education as instruments yield results exactly as expected, that is, between those shown in Tables 6 and 7.

Table 7
IV estimates using mother's education to instrument for education

Country	Males		Females	
USA	0.128	<i>0.018</i>	0.125	<i>0.019</i>
	0.076	<i>0.005</i>	0.098	<i>0.006</i>
West Germany	0.029	<i>0.009</i>	0.042	<i>0.014</i>
	0.035	<i>0.003</i>	0.046	<i>0.007</i>
Australia	0.114	<i>0.031</i>	0.129	<i>0.050</i>
	0.048	<i>0.006</i>	0.036	<i>0.008</i>
Austria	0.075	<i>0.016</i>	0.074	<i>0.019</i>
	0.037	<i>0.006</i>	0.059	<i>0.009</i>
Poland	0.074	<i>0.020</i>	0.161	<i>0.024</i>
	0.072	<i>0.007</i>	0.103	<i>0.008</i>
East Germany	0.038	<i>0.023</i>	0.049	<i>0.023</i>
	0.028	<i>0.004</i>	0.034	<i>0.006</i>
Ireland	0.130	<i>0.028</i>	0.148	<i>0.032</i>
	0.077	<i>0.016</i>	0.129	<i>0.017</i>
Hungary	0.086	<i>0.030</i>	0.075	<i>0.019</i>
	0.057	<i>0.010</i>	0.073	<i>0.010</i>
Weighted Average	0.072	<i>0.014</i>	0.103	<i>0.022</i>
	0.054	<i>0.006</i>	0.072	<i>0.008</i>

Robust standard errors are in italics. The estimating equations include year dummies, union status, marital status, age and age squared. For comparison, the corresponding OLS estimates are shown in the smaller font underneath.

correlation between reported schooling and our instruments, our IV estimates are considerably more precise than most previous IV estimates of the return to education. Thus, the previous finding that IV estimates of the rate of return to schooling are considerably higher than OLS estimates is not unique to the US and the UK. However, despite the relative precision of our IV estimates the difference between IV and OLS is significant in only two cases.¹²

The validity of our IV estimates in Tables 5–7 depends on the exclusion restriction that spouses' or parents' education does not affect wage rates. Lam and Schoeni (1993) appears to be the only previous study to have exploited assortative mating to instrument for education, and they were motivated by a desire to control for family background rather than any unobservable determinants of earnings. They conclude, albeit for a developing country, that spouse's education does indeed control for such unobservables. They note, however, that even when controls for parental education are included, spouse's education has a significant effect on wages and that including wife's schooling results in the returns to husband's schooling falling from 16.3% to 12.4%. However, this work is based on Brazilian data and the route by which wife's education influences the returns to husband's education (for example, networks or nepotism) may be less pronounced in a developed economy. The exclusion restriction that parental education has no direct effect on wages is equally, if not more, problematic. It is not difficult to think of good reasons why it should not hold. However, parental background informa-

¹² As in the OLS case, the IV-estimated rates of return are positively correlated with their standard errors. The correlation coefficients (and their *P*-values) are 0.266 (0.256), 0.53 (0.022) and 0.54 (0.030).

tion has been adopted as instruments elsewhere, and, more importantly, its validity is ultimately an empirical question.

In any case, we can test the validity of our instruments following Bound et al. (1995). These authors and others have shown that caution is needed in choosing suitable instruments. Weak instruments can induce greater bias than OLS. To test the validity of the instrument set, Bound et al. first recommend conducting an F test on the excluded instruments in the reduced-form schooling equation to ensure that education is indeed well correlated with the instruments once other controls are included. These are reported in the left-hand columns in Table 8. In every instance, the independence between schooling and the instruments is decisively rejected, particularly so for spouse's education. Our instru-

Table 8
Instrument validity tests

Country	Males		Females	
	(1)	(2)	(1)	(2)
<i>Spouse's education</i>				
USA	647.60	1.25	434.00	1.73
West Germany	129.00	1.96	119.61	7.93
Australia	223.90	0.06	157.28	3.43
Netherlands	32.37	1.55	34.40	6.07
Poland	507.83	0.00	274.50	0.02
East Germany	76.90	0.02	45.18	1.57
Italy	208.56	18.84	162.79	33.49
Ireland	93.04	0.02	114.72	2.80
Hungary	156.01	0.09	48.63	1.31
Czechoslovakia	78.19	0.49	97.44	0.37
<i>Father's education</i>				
USA	254.53	5.37	248.09	4.16
West Germany	111.23	0.30	91.89	1.51
Australia	95.92	0.95	44.88	1.46
Austria	80.55	2.29	107.35	2.52
Poland	156.09	0.25	147.37	5.30
East Germany	19.95	2.28	14.93	0.01
Ireland	59.70	3.28	23.47	3.14
Hungary	30.63	1.87	34.05	1.02
Czechoslovakia	15.95	0.20	43.76	2.77
<i>Mother's education</i>				
USA	173.17	10.79	271.17	2.45
West Germany	95.62	1.54	67.86	0.00
Australia	52.48	3.93	27.18	2.82
Austria	56.40	6.36	97.10	0.50
Poland	61.70	0.13	76.15	4.24
East Germany	9.31	1.33	12.97	0.05
Ireland	19.20	2.69	47.22	4.84
Hungary	28.36	0.80	47.55	0.02

(1) is the F test for the exclusion of the instruments in the first-stage equation. (2) is the F test for the exclusion of the instruments in the second-stage equation.

ments predict reported schooling exceedingly well, which is probably not surprising given our choices.

Bound et al. (1995) also recommend conducting an F test of the effect of the instruments on the wage residuals to ensure that the instruments are not directly correlated with the wage once the other control variables are included. These are reported in the right-hand columns of Table 8. Here the tests are less decisive. In the majority cases, the exclusion restrictions cannot be rejected, but there are numerous instances, particularly for females, where the exclusion restrictions can be rejected. That is, there are cases where the instruments do have a significant direct correlation with wages. Thus, caution should be applied to the IV estimates in Tables 5–7, particularly those for women.

The best results are those for men when using spouse's education. This instrument is not significantly correlated with wages in 9 of the 10 countries. Italy is the only case that fails the second test for instrument validity. The results for men when using father's education are almost as good. The results for the USA and Ireland (marginally) are dubious in this instance. Oddly (to us at least), mother's education appears to have more of a direct impact on wages for men. The exclusion test can be at least marginally rejected in half of the countries (USA and Ireland again, plus Australia and Austria).

The instruments are not generally as valid when applied to females' education. The exclusion test can be at least marginally rejected in almost half of the cases. Spouse's education at least marginally fails the exclusion test for females in half of the countries (Italy, especially, plus West Germany and the Netherlands, and Australia and Ireland, marginally). Father's education again fails the exclusion test for females for USA and Ireland (marginally), plus Poland and Czechoslovakia (marginally). The exclusion test for mother's education on female wages fails in Poland, Ireland and Australia (marginally).

Although there is some evidence that our instruments are not completely orthogonal to the dependent variable, we feel that this problem is not sufficient to undermine the conclusion that OLS estimates are biased downward by about 20% or more. The differences between the IV and OLS results are generally larger in the cases where the exclusion restriction is rejected, as one would expect. But the IV estimates are also noticeably higher than OLS in the cases where the exclusion restriction is not rejected. That is, there is considerable evidence that the direct correlation between the instruments and the dependent variable is not the sole reason why the IV estimates of the rate of return to education are substantially higher than OLS estimates.

7. Conclusion

We conclude that conventional OLS estimates suggest a worldwide average rate of return to schooling of just under 5% for men, and a little under 6% for women. Also, there is a great deal of variation in the return to education across countries. Moreover, much of this cross-country variation in the return to schooling defies ready explanation. IV estimates, however, indicate that these OLS estimates are biased downward by about a percentage point, possibly more. Further, there appears to be little systematic trend in return to schooling from 1985 through 1995.

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